

# Gauge-Invariant Scalarizability of Multi-Inverter Power Networks: Non-Abelian Cycle Dynamics and Connectivity–Stability Reversals

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**Abstract**—Scalar graph measures and decoupled impedance models are attractive for screening inverter-dominated power networks, but they discard matrix-valued feedback interactions created by phase-locked loops, voltage/current controls, grid-forming loops, and heterogeneous device dynamics. This paper develops a coordinate-invariant framework for deciding when such a network admits an exact scalar-channel representation, quantifying the error when it does not, and separating that model-validity question from closed-loop stability. After port reduction, a frequency-dependent matrix return graph is normalized by a spanning-tree gauge and the non-tree transports become fundamental-cycle holonomies. We prove fixed-frequency scalarizability by simultaneous diagonalization and extend it to a broadband common-cycle criterion; noncommutativity and Jordan defectiveness are distinct exact obstructions. A physical port metric yields a gauge-invariant non-Abelian defect, while a spectral-gap theorem bounds the off-diagonal residual of each chord by its commutator with a reference holonomy. Across 2,000 structured  $2 \times 2$  networks the bound has zero violations, a median slack of 6.6%, and a 95th-percentile slack of 35.9%. To isolate genuinely global information, a second 2,000-case campaign uses non-isomorphic Laplacian-cospectral graph pairs with identical local edge-coupling distributions: the scalarized spectra agree to  $7.3 \times 10^{-14}$ , yet the full MIMO models disagree on stability in 4.35% of the trials. A collective-cycle decomposition further shows that, at an unstable unit-return mode, the largest fundamental closure accounts for only 8.84% of total chord magnitude and the effective number of participating closures is 28.9. Finally, an AC-consistent IEEE 39-bus reduced SG/GFL/GFM study is used as a falsification-oriented engineering test. Under nominal PLL tuning, 180 candidate line additions produce no instability. In a deliberately low-damped PLL region, line 25–38 improves generalized-Fiedler connectivity while the full coupled mode crosses into instability and a decoupled model misses the event. Frequency-resolved cycle diagnostics reveal a non-Abelian feature near the return-ratio critical band while the scalarization residual remains broadband; the reversal occupies a finite controller/line-strength region, changes sign with operating point, is absent in 720 sampled GFM upgrade trials, and is repaired by retuning one attributed PLL without removing the line. The claim is deliberately narrow: cycle algebra certifies scalar-model validity; stability remains a property of the full closed-loop matrix graph.

**Index Terms**—Inverter-dominated power systems, MIMO impedance, graph theory, holonomy, quiver representations, non-Abelian coupling, scalarization, generalized Fiedler vector, PLL, small-signal stability, Braess effect.

Manuscript draft, August 2026. The IEEE 39-bus study in this draft uses an AC-consistent reduced dynamic surrogate; a full-order EMT/Simulink/ANDES replication is identified explicitly as the remaining external validation gate.

## I. INTRODUCTION

MODERN power networks combine synchronous machines, grid-following (GFL) inverters, grid-forming (GFM) inverters, dynamic loads, and controls whose terminal dynamics are naturally multivariable. In this setting, a line is not merely a scalar weight connecting two otherwise identical oscillators. A perturbation can rotate, amplify, and mix voltage-angle and power channels as it traverses the network. Yet many planning and screening tools remain scalar: Laplacian connectivity, effective resistance, short-circuit-ratio families, or decoupled impedance approximations.

This mismatch matters for two reasons. First, the recently proposed generalized Fiedler vector (GFV) combines network connectivity with nodal inertia and provides a useful notion of *dynamic connectivity* [1]. Second, converter impedance studies have long shown that dropping coupling terms can lead to inaccurate eigenloci or even incorrect stability conclusions when mirror-frequency or  $dq$  coupling is material [6], [7]. These two facts create a practical gap: an upgrade can look favorable to a scalar connectivity metric while the closed-loop MIMO dynamics react in another direction.

Complex-frequency formulations provide an increasingly useful language for heterogeneous power-system dynamics [2], [3], [5]; however, complex frequency and GFV are not the contribution pursued here. Likewise, Braess-type stability degradation after adding a coupling is already known for oscillator networks and power grids [10]. The question addressed here is more specific:

*When does a matrix-valued power-network model admit an exact scalar-channel representation, and what closed-loop information is missing when such a representation does not exist?*

The answer developed in this paper uses cycle algebra. Each local interconnection is treated as a linear map between port spaces. Local state-coordinate changes act as gauge transformations. A spanning-tree gauge removes all tree transports, leaving the fundamental-cycle return maps—their *holonomies*—as the nontrivial global objects. This leads to four contributions.

**C1. Exact scalarizability.** A tree-gauge theorem characterizes when a connected matrix-valued network can be transformed into independent scalar channels: precisely

when its fundamental-cycle holonomies are simultaneously diagonalizable.

- C2. Gauge-aware obstruction and topology-only nonidentifiability.** A positive-definite physical port metric yields a coordinate-invariant non-Abelian commutator defect and a spectral-gap error certificate. In addition to the 2,000-case bound test, 2,000 non-isomorphic Laplacian-cospectral graph-pair trials preserve the scalar spectrum and local coupling statistics while producing different full-MIMO stability verdicts in 4.35% of cases.
- C3. Collective feedback.** A counterexample shows that global unit-return resonance does not require any single fundamental cycle to be individually close to unit return. An exact unit-return decomposition on the IEEE 39-bus stress case quantifies the distributed closure: the largest chord carries only 8.84% of total chord magnitude and the effective number of participating closures is 28.9.
- C4. Connectivity–stability nonordering, robustness, and mitigation.** An exact minimal MIMO counterexample and an AC-consistent IEEE 39-bus study show that improved scalar connectivity need not improve the full coupled spectrum. The reversal is mapped over line strength, PLL tuning, IBR penetration, and loading; it is controller- and operating-point-dependent, and targeted PLL retuning restores the margin without undoing the network upgrade.

The paper deliberately separates *scalarizability* from *stability*. Non-Abelianity is not proposed as a stability margin. Rather, it certifies when a scalar reduction cannot retain the complete matrix-valued feedback structure.

## II. RELATED WORK AND POSITIONING

### A. Scalar connectivity and complex frequency

The GFV recently proposed by Pinheiro and Dotta generalizes Fiedler connectivity by incorporating nodal inertia [1]. In parallel, local and system-wide complex-frequency formulations provide a device-agnostic description of voltage magnitude and angle dynamics [2], [3]. Büttner and Hellmann further introduced complex couplings as adaptive dynamical-network variables [5]. These developments provide useful baselines and future observables, but none addresses the exact reducibility of a heterogeneous matrix-valued interconnection into scalar channels.

### B. Coupled versus decoupled converter models

Matrix impedance and generalized Nyquist approaches explicitly retain  $dq$ /sequence coupling. Rygg *et al.* showed that decoupled impedance models are exact only under specific coupling conditions and quantified eigenlocus error [6]. Xin *et al.* likewise showed that neglected coupling can produce incorrect conclusions and proposed a generalized-impedance treatment [7]. The present work is complementary: instead of assessing coupling at one converter, it asks whether the *entire network* admits one global basis in which every interconnection channel decouples.

### C. Cycles, holonomy, and quiver determinants

Cycle-level topological constraints have recently appeared in synchronization through twisted Laplacians and  $U(1)$  holonomy [9]. More generally, Kassel and Lévy derived determinant identities for linear quiver representations in terms of traces of cycle holonomies, including cycle-rooted spanning-forest interpretations for graph vector bundles [8]. We do not claim these algebraic results. We use their perspective to motivate a power-system interpretation: matrix-valued feedback cycles are the coordinate-invariant structures left after trivializing a spanning tree.

### D. Braess effects and stability nonordering

Adding a network coupling can either increase or decrease synchronous-state stability, including in power grids [10]. Our IEEE 39-bus experiment should therefore not be read as a claim that Braess behavior itself is new. The new target is the mismatch between scalar connectivity and a coupled converter mode, together with a scalarizability diagnosis and a controller-retuning remedy.

## III. MATRIX-VALUED CLOSED-LOOP NETWORK

### A. Port reduction and self-energy interpretation

Let  $z$  denote retained terminal variables and  $x_h$  internal machine, converter, filter, and controller states. Linearizing a DAE around an AC solution and eliminating  $x_h$  by a Schur complement yields a frequency-dependent port equation

$$F(s)z = 0, \quad (1)$$

where  $F(s)$  contains the network and the dynamic self-energy of eliminated states. A convenient block splitting is

$$F(s) = D(s) - E(s), \quad (2)$$

with block-diagonal local dynamics  $D(s) = \text{diag}(D_1, \dots, D_n)$  and off-diagonal interconnection blocks  $E_{ij}(s)$ . Whenever  $D(s)$  is invertible,

$$F(s) = D(s)[I - T(s)], \quad T(s) = D^{-1}(s)E(s), \quad (3)$$

so

$$\det F(s) = \det D(s) \det[I - T(s)]. \quad (4)$$

A collective network mode therefore satisfies

$$1 \in \sigma[T(s)]. \quad (5)$$

The block  $T_{ij}(s)$  maps the port space of node  $j$  to that of node  $i$  and embeds inertia, damping, controller memory, and network transfer. This is the point at which the earlier “feedback rheology” interpretation becomes useful: frequency-dependent local dynamics change the transport around network cycles rather than simply adding one scalar damping coefficient.

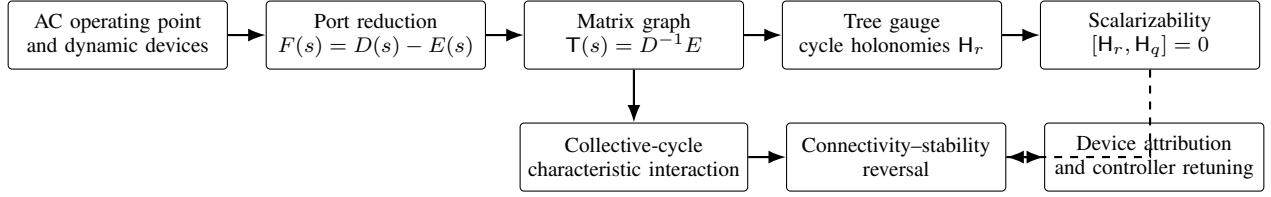


Fig. 1. Paper logic. The contribution is not a new scalar strength metric. The port-reduced MIMO network is treated as a matrix graph; cycle algebra decides exact scalarizability, while the full closed-loop model decides stability.

### B. Gauge transformations

Let each port space be  $V_i \simeq \mathbb{C}^d$  and let  $x_i = Q_i \tilde{x}_i$  with  $Q_i \in \text{GL}(d, \mathbb{C})$ . Then

$$T_{ij} \mapsto \tilde{T}_{ij} = Q_i^{-1} T_{ij} Q_j. \quad (6)$$

The local realization can therefore change while the underlying physical interconnection does not. A useful network descriptor should either be invariant under (6) or carry the physical metric required to transform covariantly.

**Definition 1** (Cycle holonomy). For an oriented closed walk  $C = (i_0, i_1, \dots, i_r = i_0)$ , define

$$H_C = T_{i_0 i_{r-1}} \cdots T_{i_2 i_1} T_{i_1 i_0}. \quad (7)$$

Under (6),  $H_C \mapsto Q_{i_0}^{-1} H_C Q_{i_0}$ . Hence its eigenvalues, trace, determinant, and characteristic polynomial are gauge-invariant.

## IV. EXACT SCALARIZABILITY

### A. Single-cycle reduction and tree gauge

The simplest consistency check is a directed cycle with block transports  $T_1, \dots, T_r$ . Let  $T_C$  be the corresponding block cyclic return operator and  $H_C = T_r \cdots T_1$ .

**Lemma 1** (Single-cycle determinant reduction). *For a pure directed cycle of equal block dimension,*

$$\det(I - T_C) = \det(I - H_C). \quad (8)$$

*Proof.* Successive block Gaussian elimination removes the tree-like chain blocks and leaves the Schur complement  $I - T_r \cdots T_1$ . Equivalently, the nonzero closed-walk contribution of the block permutation is the cycle product  $H_C$ .  $\square$

A 3-node,  $2 \times 2$  numerical check gives an absolute determinant mismatch of  $3.5 \times 10^{-16}$ . Under strongly non-unitary local coordinate changes, the spectrum, trace, and determinant of  $H_C$  remain invariant to machine precision. In contrast, the Euclidean quantity  $\sigma_{\min}(I - H_C)$  changed from 0.9026 to 0.5132 under a pure coordinate transformation. This motivates the physical metric introduced below.

### B. Tree gauge

Choose a spanning tree  $\mathcal{T}_G$  and a root. Assume the tree transports are invertible. Recursively select the  $Q_i$  so that every tree transport becomes the identity. The remaining  $\beta = m - n + 1$  chord matrices are exactly the fundamental-cycle holonomies in this tree trivialization.

**Theorem 1** (Tree-gauge scalarizability). *For a connected matrix-valued network with equal port dimension  $d$  and invertible tree transports, the following statements are equivalent:*

- (i) *there exist local coordinate transformations that make every edge transport diagonal;*
- (ii) *in a tree gauge, all fundamental-cycle holonomies  $H_1, \dots, H_\beta$  are simultaneously diagonalizable.*

*If the  $H_r$  are individually diagonalizable over  $\mathbb{C}$ , (ii) is equivalent to pairwise commutation  $[H_r, H_q] = 0$  for all  $r, q$ .*

*Proof.* In a tree gauge all tree transports are  $I$ . Any further collection of local transformations that preserves all tree identities must satisfy  $S_i = S_j$  on every tree edge; connectedness therefore implies a single similarity  $S$  at all vertices. The only nontrivial transports are the chord matrices  $H_r$ . Hence all edges can be diagonalized if and only if one  $S$  diagonalizes all  $H_r$ . The final statement is the standard simultaneous-diagonalization criterion for a commuting family of diagonalizable complex matrices [15].  $\square$

**Remark 1.** The theorem is a *structural* exactness result. It does not state that a commuting network is stable, only that it admits a representation as  $d$  independent scalar graph channels.

Because  $T_{ij} = T_{ij}(s)$ , pointwise scalarizability does not by itself guarantee one physically useful scalar decomposition over a frequency band. Let the spanning tree be fixed and, at each  $\omega$ , perform the corresponding tree normalization. The remaining root-fiber holonomies are denoted  $H_r(j\omega)$ .

**Definition 2** (Broadband common-cycle scalarizability). A matrix network is broadband common-cycle scalarizable on  $\Omega$  if there exists one similarity  $S$ , independent of  $\omega$ , such that  $S^{-1}H_r(j\omega)S$  is diagonal for every fundamental cycle  $r$  and every  $\omega \in \Omega$ .

**Theorem 2** (Broadband common-cycle criterion). *Assume that every member of the family  $\{H_r(j\omega) : r = 1, \dots, \beta, \omega \in \Omega\}$  is diagonalizable over  $\mathbb{C}$ . Then broadband common-cycle scalarizability holds if and only if*

$$[H_r(j\omega), H_q(j\nu)] = 0 \quad \forall r, q, \omega, \nu \in \Omega. \quad (9)$$

*Proof.* The statement is simultaneous diagonalization of the entire frequency-indexed family. A simultaneously diagonal family commutes. Conversely, a commuting family of diagonalizable complex matrices is simultaneously diagonalizable [15].  $\square$

The diagonalizability hypothesis is essential. Pairwise commutation alone cannot detect a Jordan obstruction. For example,

$$J_1 = \begin{bmatrix} 1 & 1 \\ 0 & 1 \end{bmatrix}, \quad J_2 = \begin{bmatrix} 2 & 1 \\ 0 & 2 \end{bmatrix} \quad (10)$$

commute but neither is diagonalizable. We therefore distinguish the *non-Abelian obstruction* from the *Jordan obstruction*. One exact diagnostic is

$$\mathcal{J}(\omega) = \max_r \left[ d - \sum_{\lambda \in \sigma(H_r)} \dim \ker(H_r - \lambda I) \right], \quad (11)$$

which vanishes exactly when every  $H_r$  is diagonalizable. Numerically, eigenvector conditioning should accompany  $\mathcal{J}$  to detect near-defectiveness. Theorems 1–2 are structural statements, not stability conditions.

### C. Gauge-invariant non-Abelian defect

A Euclidean Frobenius norm of  $[H_r, H_q]$  is not invariant under non-unitary changes of local coordinates. Let  $G \succ 0$  be a physical quadratic port metric, transforming as  $G \mapsto Q^* G Q$ . Define

$$\|A\|_G^2 = \text{tr}(G^{-1} A^* G A) \quad (12)$$

and

$$\mathfrak{N}_{rq} = \frac{\|H_r H_q - H_q H_r\|_G}{\|H_r H_q\|_G + \|H_q H_r\|_G}. \quad (13)$$

**Proposition 1** (Gauge invariance). *The norm (12) and defect (13) are invariant under  $A \mapsto Q^{-1} A Q$ ,  $G \mapsto Q^* G Q$ .*

*Proof.* Substitution into (12), cyclicity of the trace, and  $Q^{-1}(Q^*)^{-1} = (Q^* Q)^{-1}$  cancel all basis factors. Applying the same identity to the commutator and denominator terms yields (13) unchanged.  $\square$

A strongly non-unitary numerical gauge transformation changed  $\mathfrak{N}$  by  $4.2 \times 10^{-17}$ , whereas an unweighted Euclidean singular-value quantity changed materially. This correction is important: gauge invariance requires carrying the physical metric.

### D. A quantitative spectral-gap certificate

Gauge-invariant commutators indicate whether exact scalarization is possible, but a screening tool also needs a quantitative statement. Let  $G = W^* W \succ 0$  and whiten the physical metric by  $\hat{H} = W H W^{-1}$ . Choose a reference holonomy  $\hat{H}_0$  with simple spectrum and write

$$S^{-1} \hat{H}_0 S = \Lambda = \text{diag}(\lambda_1, \dots, \lambda_d), \quad (14)$$

$$\delta_0 = \min_{i \neq j} |\lambda_i - \lambda_j| > 0. \quad (15)$$

For another holonomy  $\hat{H}_q$ , let  $B_q = S^{-1} \hat{H}_q S$  and let  $\text{off}(B_q)$  denote its off-diagonal part.

**Theorem 3** (Commutator-to-scalarization bound). *Under the assumptions above,*

$$\|\text{off}(B_q)\|_F \leq \frac{\kappa_2(S)}{\delta_0} \|[\hat{H}_0, \hat{H}_q]\|_F. \quad (16)$$

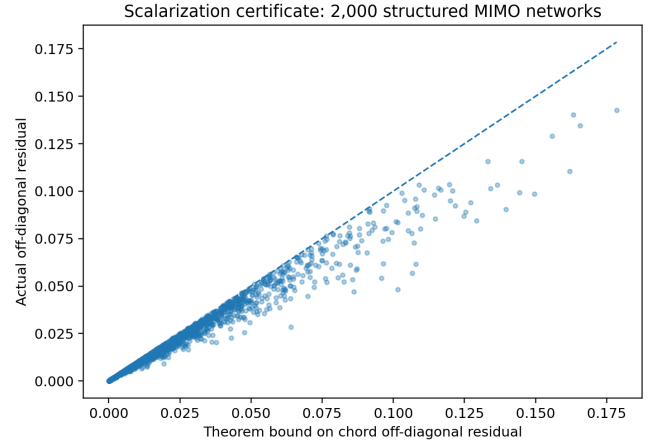


Fig. 2. Quantitative scalarization certificate for 2,000 structured MIMO networks. Every realization lies below the theorem line (dashed). The median bound/actual ratio is 1.066 and the 95th percentile is 1.359.

*Proof.* In the reference eigenbasis,  $[\Lambda, B_q]_{ij} = (\lambda_i - \lambda_j)(B_q)_{ij}$ . Hence  $\|\text{off}(B_q)\|_F \leq \delta_0^{-1} \|[\Lambda, B_q]\|_F$ . Because  $[\Lambda, B_q] = S^{-1}[\hat{H}_0, \hat{H}_q]S$ , submultiplicativity gives (16).  $\square$

The bound makes two failure mechanisms explicit: large cycle noncommutativity and a small reference spectral gap. The factor  $\kappa_2(S)$  additionally penalizes a nearly defective reference basis. Thus the practical scalarization certificate combines commutator size, eigenvalue separation, and modal conditioning rather than treating  $\mathfrak{N}$  alone as sufficient.

## V. STRUCTURED MONTE CARLO SCALARIZATION TEST

### A. Protocol

We generated 2,000 four-node networks with a directed spanning tree  $0 \rightarrow 1 \rightarrow 2 \rightarrow 3$  and chords  $2 \rightarrow 0$  and  $3 \rightarrow 0$ . Each  $2 \times 2$  edge transport was a rotated anisotropic positive-definite matrix. After tree gauge and metric whitening, the first chord was used as the reference  $\hat{H}_0$ ; the actual residual was the off-diagonal Frobenius norm of the second chord in the reference eigenbasis. The theorem bound in (16) was evaluated for every realization. In parallel, the full block network and the chord-diagonalized approximation were compared spectrally.

### B. Results

There were no violations of (16) in 2,000 realizations. The median bound-to-actual ratio was 1.0659, and the 95th percentile was 1.3589. Thus, for this structured ensemble, the elementary spectral-gap certificate is substantially sharper than a subsequent Bauer–Fike bound on the full spectral shift, whose median slack exceeded  $800\times$  and is therefore not used as the practical certificate.

The energy-metric non-Abelian defect remains informative about the final scalar-network error: Spearman  $\rho_s = 0.8951$  and Pearson  $r = 0.5306$ . The upper-quartile median scalarization error is approximately 394 times the lower-quartile median. Finally, 200 exactly commuting cases under arbitrary local gauges remained exact to machine precision, with maximum scalarization error  $4.8 \times 10^{-15}$ .

TABLE I  
SCALARIZABILITY VALIDATION SUMMARY

| Quantity                                       | Result                |
|------------------------------------------------|-----------------------|
| Structured MIMO realizations                   | 2000                  |
| Violations of (16)                             | 0                     |
| Median bound/actual residual                   | 1.0659                |
| 95th-pct. bound/actual residual                | 1.3589                |
| Spearman $\rho_s(\mathfrak{N}, \epsilon_{sc})$ | 0.8951                |
| Exactly commuting gauge tests                  | 200                   |
| Max. commuting-case error                      | $4.8 \times 10^{-15}$ |
| Gauge-change error in $\mathfrak{N}$           | $4.2 \times 10^{-17}$ |

The interpretation remains deliberately limited: the theorem and  $\mathfrak{N}$  quantify *model reducibility*. They do not order closed-loop stability.

## VI. TOPOLOGY-ONLY NONIDENTIFIABILITY AND SCREENING ABLATION

The preceding Monte Carlo varies local matrix transports and therefore does not, by itself, prove that global cycle organization contains information unavailable to a local coupling index. We therefore constructed a stricter falsification test based on non-isomorphic *Laplacian-cospectral* graph pairs. Within each pair, the two networks have the same number of nodes and edges, the same degree sequence, the same scalar Laplacian spectrum, the same nodal matrix  $A$ , and the same multiset of local  $2 \times 2$  edge blocks. Every local edge-coupling magnitude is therefore unchanged; only the way those matrix blocks close through cycles is altered.

For one exact seven-node pair, the scalarized model predicts the identical spectral abscissa  $-0.44309$  for both networks, with Laplacian-spectrum mismatch below  $2.8 \times 10^{-15}$ . The full MIMO network A has  $\alpha = -0.36811$ , whereas network B has  $\alpha = +0.02838$ . Thus two networks that are indistinguishable to the selected scalar spectrum and local coupling statistic have opposite stability verdicts.

The experiment was repeated over 2,000 randomly parameterized cospectral graph pairs. The largest scalar-spectrum discrepancy was  $7.27 \times 10^{-14}$ , while the median full-MIMO spectral-abscissa gap was 0.0124 and the 95th percentile was 0.1453. Most importantly, 4.35% of paired trials produced opposite full-MIMO stability verdicts. This establishes a nonidentifiability result stronger than correlation: neither the scalar Laplacian spectrum nor an unchanged local coupling distribution is a sufficient statistic for the matrix-network stability verdict.

A complementary fixed-topology ablation prevents the opposite overclaim. On 1,000 randomized IEEE 39-bus stress configurations, a simple local  $dq$  coupling feature was a better standalone empirical classifier of an “unsafe scalarization” event than the cycle defect: repeated-cross-validation AUC was  $0.726 \pm 0.041$  for local coupling and  $0.673 \pm 0.047$  for the non-Abelian cycle feature, while GFV and standard Fiedler were near chance for this classification task. Table II summarizes the result. Consequently,  $\mathfrak{N}$  is not proposed as a generic instability classifier; its value is structural certification and exact counterexamples that local metrics cannot provide.

TABLE II  
REPEATED-CV AUC FOR UNSAFE-SCALARIZATION SCREENING

| Feature set                  | ROC AUC           |
|------------------------------|-------------------|
| Local $dq$ coupling          | $0.726 \pm 0.041$ |
| GFV + local coupling         | $0.722 \pm 0.039$ |
| GFV + local + cycles         | $0.712 \pm 0.039$ |
| Non-Abelian cycles           | $0.673 \pm 0.047$ |
| Cycle scalarization residual | $0.651 \pm 0.039$ |
| Standard Fiedler             | $0.505 \pm 0.042$ |
| GFV                          | $0.466 \pm 0.047$ |

## VII. COLLECTIVE-CYCLE FEEDBACK

A tempting cycle criterion would search for one fundamental cycle with  $1 \in \sigma(H_C)$ . The following counterexample shows why that is insufficient.

**Proposition 2** (No single resonant cycle is necessary). *There exist matrix networks for which  $1 \in \sigma(T)$  while  $1 \notin \sigma(H_r)$  for every fundamental-cycle holonomy.*

*Constructive numerical proof.* In a four-port, two-chord tree gauge, the global return operator satisfies  $1 \in \sigma(T)$  to machine precision. However,

$$\sigma(H_1) = \{0.51884, 0.24877\}, \quad (17)$$

$$\sigma(H_2) = \{0.50223, 0.18559\}, \quad (18)$$

so the nearest individual cycle is more than 0.48 from unit return. Nevertheless, sensitivities of the global unit eigenvalue to the two chord gains are 0.14528 and 0.14104, respectively. Thus both cycles participate in the global condition although neither is individually resonant.  $\square$

This motivates a *collective-cycle* characteristic description. Recent trace identities for quiver representations express suitable block determinants through traces of products along cycles, and twisted-Laplacian determinants through cycle-rooted spanning forests [8]. We use this machinery as mathematical infrastructure rather than claim it as new.

A directly useful power-system quantity follows from eigenvalue perturbation of the return graph. Suppose  $T(s_k)x_k = x_k$ ,  $y_k^* T(s_k) = y_k^*$  and  $y_k^* x_k = 1$ . Let a tree-gauged chord block  $H_a$  be scaled by a scalar  $\gamma_a$ .

**Proposition 3** (Gauge-invariant chord sensitivity). *For a simple unit-return eigenvalue,*

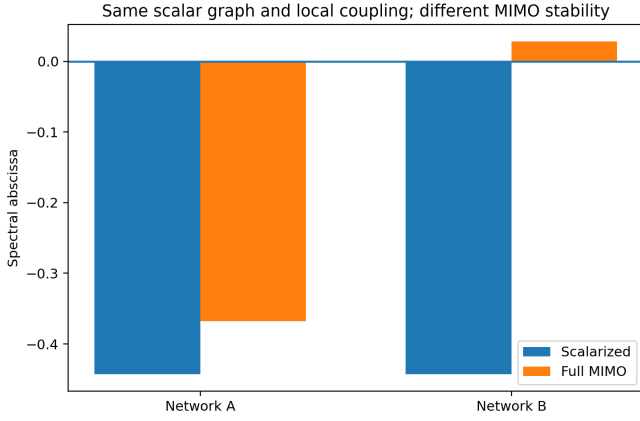
$$\frac{d\lambda_T}{d\gamma_a} = y_{k,i}^* H_a x_{k,j}, \quad (19)$$

*and the corresponding closed-loop mode sensitivity is*

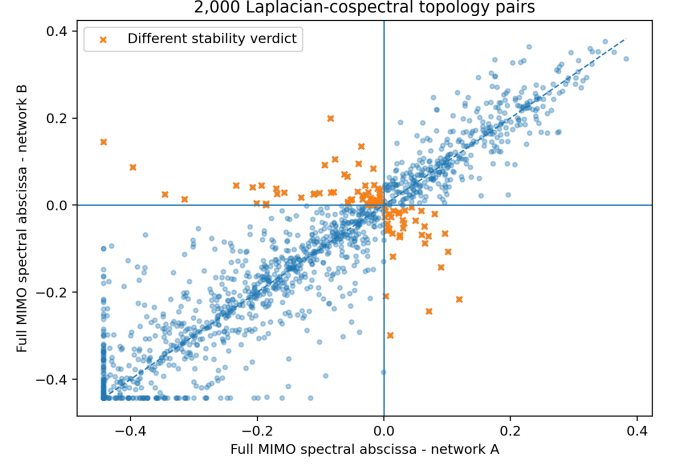
$$\frac{ds_k}{d\gamma_a} = -\frac{y_{k,i}^* H_a x_{k,j}}{y_k^* T(s_k) x_k}. \quad (20)$$

*Both expressions are invariant under local gauge changes when the left/right vectors transform contragrediently.*

*Proof.* Differentiate  $y_k^*(T - \lambda_T I)x_k = 0$  with respect to  $\gamma_a$  to obtain (19). At a closed-loop mode  $\lambda_T(s_k) = 1$ , implicit differentiation of  $\lambda_T(s_k, \gamma_a) = 1$  yields (20).  $\square$



(a) Exact iso-local, Laplacian-cospectral counterexample.



(b) Full-MIMO outcomes over 2,000 cospectral topology-pair trials. Crosses denote opposite stability verdicts.

Fig. 3. Topology-only falsification of scalar/local sufficiency. The scalarized spectra are identical within each pair, and each pair uses the same local edge-coupling distribution; only the global arrangement of matrix-valued couplings differs.

For a complete characteristic attribution, Kassel–Lévy-type trace/CRSF identities motivate

$$P(s) = \det F(s) = \sum_{\mathcal{F}} \Psi_{\mathcal{F}}(s), \quad (21)$$

but Proposition 3 already provides a scalable chord-level diagnostic without enumerating all cycle-rooted spanning forests.

The IEEE 39-bus unit-return mode allows the collective statement to be quantified exactly. With left/right normalization  $y^*x = 1$ , the directed block contributions satisfy

$$1 = y^*Tx = \sum_{i \neq j} y_i^* T_{ij} x_j \quad (22)$$

to  $9 \times 10^{-16}$  in real part and  $8 \times 10^{-16}$  in imaginary part. Among the 81 non-tree closure channels, the Herfindahl concentration of absolute contribution is only 0.0346, corresponding to an effective closure count of

$$N_{\text{eff}} = \left( \sum_a p_a^2 \right)^{-1} = 28.9. \quad (23)$$

The single largest chord contributes 8.84% of total chord magnitude and the top five together only 34.6%. Figure 4 shows the signed exact decomposition. This is direct evidence that “collective cycle” is not merely descriptive language for this mode.

## VIII. CONNECTIVITY DOES NOT ORDER MIMO STABILITY

### A. Exact minimal counterexample

Consider

$$\dot{x} = (I \otimes A - L \otimes C)x \quad (24)$$

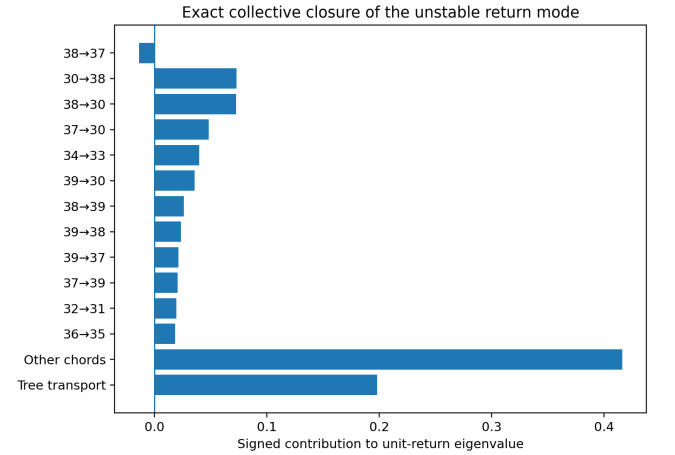


Fig. 4. Exact signed decomposition of the post-upgrade unit-return eigenvalue into effective cycle-closing channels after tree gauge. No individual closure dominates the collective condition.

for two nodes joined by an edge of weight  $w$ , so  $\mu_2(L) = 2w$ . Let

$$A = \begin{bmatrix} 0.719614 & -1.819392 \\ 0.794706 & -1.686641 \end{bmatrix}, \quad C = \begin{bmatrix} 0.283668 & -0.501948 \\ 0.467149 & 0.202568 \end{bmatrix}. \quad (25)$$

Both  $A$  and  $-C$  are Hurwitz. The relative mode is  $A - \mu C$  and

$$\det(A - \mu C) = 0.2919466\mu^2 - 0.9161545\mu + 0.2321517. \quad (26)$$

The two positive roots are

$$\mu_c = 0.2780313, \quad \mu_2^{(+)} = 2.8600578, \quad (27)$$

or, in edge weight,

$$w_c = 0.1390156, \quad w_2 = 1.4300289. \quad (28)$$

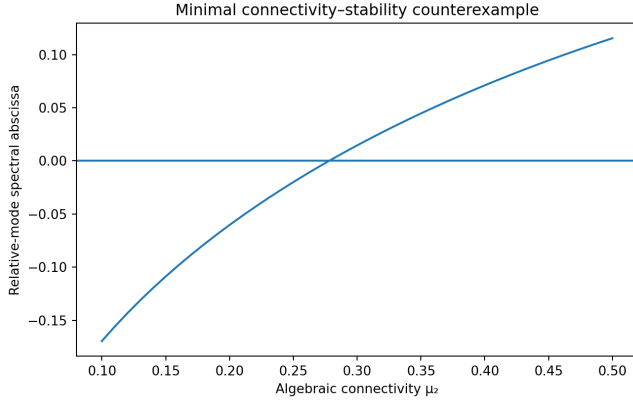


Fig. 5. Exact minimal counterexample. Increasing algebraic connectivity crosses the closed-loop stability boundary at  $\mu_c = 0.2780313$ . The purpose is to disprove a universal connectivity ordering, not to claim a new diffusion-driven instability mechanism.

The trace remains negative, while the determinant is negative between the two roots. Hence one relative-mode eigenvalue is positive for  $w_c < w < w_2$ . The first crossing at  $w_c$  is the connectivity-increase destabilization used in the argument.

**Theorem 4** (Connectivity-stability nonordering). *There is no universal ordering in which increasing scalar algebraic connectivity  $\mu_2$  must improve the spectral abscissa of a matrix-coupled network. In particular, the system (24)–(25) is stable for  $0 < w < w_c$  and unstable on the nonempty interval  $w_c < w < w_2$ , with  $w_2 = 1.4300289$ , although  $\mu_2 = 2w$  increases monotonically throughout the interval.*

Related nonmonotone phenomena are known from master-stability and diffusion-driven systems [11]; the contribution here is to use the counterexample as the minimal analytical support for the converter-network screening problem studied next.

## IX. AC-CONSISTENT IEEE 39-BUS VALIDATION

### A. Electrical benchmark and device placement

The solved MATPOWER New England 39-bus case is used as the base AC operating point [13]. For every candidate new line, the nonlinear AC power flow is re-solved, the full  $P, Q$  injection Jacobian with respect to  $(\theta, V)$  is recomputed, and the 29 load buses are eliminated by a Schur complement. The remaining ten ports correspond to the generator buses.

Four converter placements follow the commonly used low-inertia Config.-II replacement pattern, corresponding here to buses 30, 34, 37, and 38. The public EPFL benchmark provides full-replica configurations in which four synchronous machines are replaced by type-III wind plants [14]. Our present device equations are reduced surrogates rather than the full EPFL Simulink models; this distinction is maintained throughout the paper.

### B. Reduced dynamic models

The synchronous generators use a classical source-behind-reactance representation with swing dynamics. Each GFL

TABLE III  
IEEE 39-BUS DYNAMIC SCREENING CASES

| Case              | PLL $K_p$               | PLL $K_i$ |
|-------------------|-------------------------|-----------|
| Nominal           | 25                      | 400       |
| Low-damped stress | 1                       | 400       |
| Mitigated bus 30  | 3 (bus 30), 1 otherwise | 400       |

device uses a PI PLL and first-order active/reactive current-command surrogates while retaining first-order  $P/Q$  cross-coupling caused by PLL angle error. A filtered  $P$ - $f/Q$ - $V$  GFM helper model is included in the reproducibility code, although the strongest reversal below uses GFL units. The resulting reduced model is physically structured but should not be confused with a vendor-grade EMT realization. The retained core dynamics are, schematically,

$$\dot{\delta}_g = \omega_b \Delta \omega_g, \quad (29)$$

$$M_g \dot{\Delta \omega}_g = -D_g \Delta \omega_g - \Delta P_{e,g}, \quad (30)$$

for each SG, and

$$\dot{\hat{\theta}}_i = -K_{p,i} V_i (\hat{\theta}_i - \theta_i) + K_{i,i} \xi_i, \quad (31)$$

$$\dot{\xi}_i = -V_i (\hat{\theta}_i - \theta_i), \quad (32)$$

for each GFL PLL, augmented by first-order  $P/Q$  command states. Linearizing the terminal rotation gives cross terms proportional to the operating-point  $P_i$  and  $Q_i$ ; these are retained in the full model and removed in the deliberately decoupled baseline. A GFM helper uses filtered  $P$ - $f$  and  $Q$ - $V$  droop states.

For graph baselines, a synchronizing Laplacian

$$K = \sum_{(i,j)} V_i V_j B_{ij} \cos(\delta_i - \delta_j) b_{ij} b_{ij}^T \quad (33)$$

is Kron-reduced to the ten retained ports. Standard Fiedler connectivity is the second eigenvalue of  $K_r$ . GFV connectivity is the second generalized eigenvalue

$$K_r \phi = \mu M \phi, \quad (34)$$

with synchronous-machine inertia and small converter inertia surrogates.

### C. Nominal-tuning falsification gate

With  $K_p = 25$ ,  $K_i = 400$ , the baseline critical mode is

$$-0.15503 + j7.07240 \text{ s}^{-1}. \quad (35)$$

Among 180 candidate line additions at graph distance two or three, 173 increase GFV and 73 both increase GFV and move the critical real part to the right. None becomes unstable. This is an important negative result: the paper does not claim that ordinary line reinforcement generically destabilizes a well-tuned converter network.

IEEE 39-bus study topology and converter placement

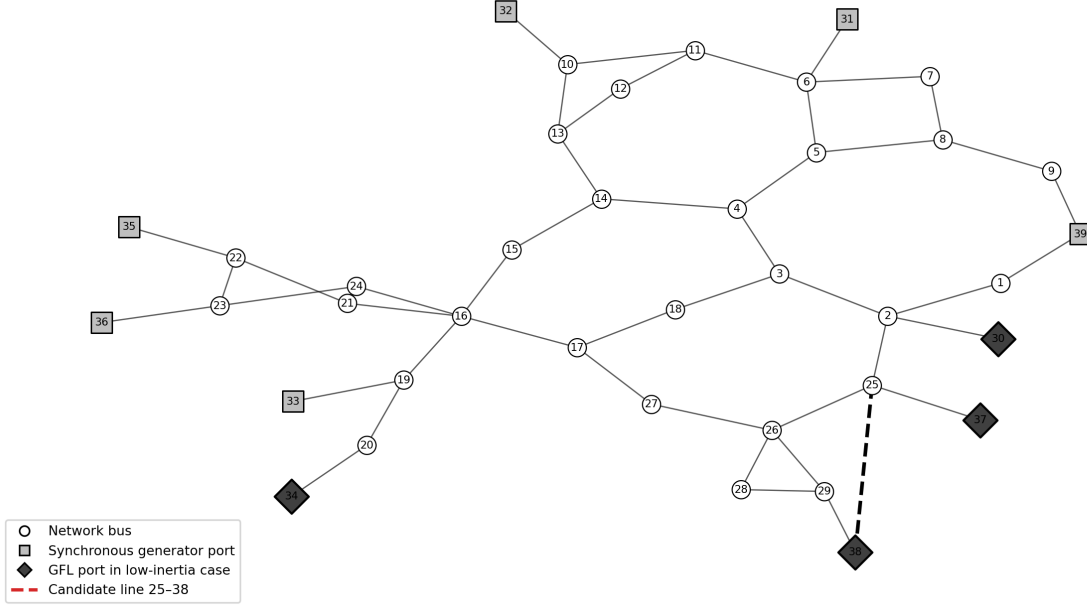


Fig. 6. IEEE 39-bus study topology. Generator buses are retained as dynamic ports after load-bus Schur reduction. The low-inertia GFL placements are buses 30, 34, 37, and 38; line 25–38 is the principal candidate intervention. The full benchmark topology and operating point are taken from MATPOWER, while converter dynamics are reduced surrogates.

#### D. Low-damped PLL stress gate

We next set  $K_p = 1$ ,  $K_i = 400$  on the four GFL units. In the local normalized PLL surrogate this corresponds to approximately  $\omega_n = 20.5$  rad/s (3.26 Hz) and  $\zeta \approx 0.026$ . The baseline is still stable:

$$s_0 = -0.01507 + j17.22591 \text{ s}^{-1}. \quad (36)$$

Of the same 180 candidate lines, four both increase GFV and produce an unstable full coupled mode. Table IV lists them.

### X. CASE STUDY: LINE 25–38

#### A. Connectivity improves while the coupled mode destabilizes

For the full line addition,

$$\mu_2^{\text{GFV}} : 0.2349169 \rightarrow 0.2352597 \quad (+0.1459\%), \quad (37)$$

$$\lambda_2(K_r) : 12.08886 \rightarrow 12.12489 \quad (+0.2980\%). \quad (38)$$

At the same time, the full coupled critical mode moves to

$$s_{\text{crit}} = +0.25019 + j19.77892 \text{ s}^{-1}. \quad (39)$$

The stability crossing occurs already at  $\alpha_c = 0.21298$  of the candidate line admittance; at the crossing, GFV remains 0.1361% above its pre-upgrade value and standard Fiedler 0.3356% above its pre-upgrade value.

#### B. Decoupled model misses the event

A decoupled approximation removes the  $P$ - $V$ ,  $Q$ - $\theta$ , and PLL-induced  $P/Q$  rotation-coupling terms. It predicts

$$-0.15627 + j7.30340 \quad \text{before the line}, \quad (40)$$

$$-0.15625 + j7.30515 \quad \text{after the line}. \quad (41)$$

Thus the scalarized/decoupled representation indicates essentially no adverse change while the full coupled model moves from stable to unstable. This is the main engineering use case for the scalarizability viewpoint: the error is not that a scalar graph metric is numerically poor, but that a scalar representation has omitted a coupled feedback channel.

#### C. Cycle-algebra diagnosis of the IEEE 39-bus mode

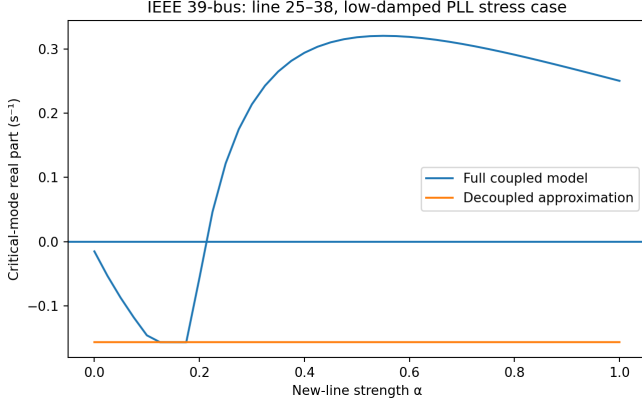
The converter-port model was next evaluated directly as the return graph  $T(s) = D^{-1}(s)E(s)$ . A maximum-weight generator-port spanning tree was fixed and all remaining directed port couplings were treated as chord closures in the tree gauge. A well-conditioned reference holonomy was used to construct the best compatible cycle-scalarized basis in the PLL band.

At the exact full-model modes, the cycle off-diagonal residual changes from 0.0673 before the upgrade to 0.1733 at the line-strength stability crossing and 0.1560 after the full line. More importantly, the full return graph satisfies the unit-return condition to  $5.4 \times 10^{-15}$  at the crossing, while the cycle-scalarized model remains 0.296 away from unit return. Thus the scalarization failure is not only visible as a generic commutator: it prevents the reduced cycle channels from reproducing the actual closed-loop mode.

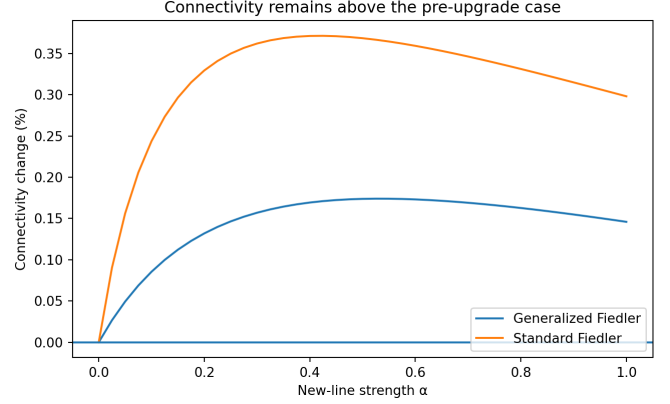
A denser frequency sweep provides the missing mechanistic view. At the stability crossing the minimum imaginary-axis return-ratio distance occurs near 19.25 rad/s, where the compatible scalarization residual is 0.169 and the physical-basis local coupling ratio is 0.870. Figure 9 normalizes three independent diagnostics only for visual comparison: a similarity-invariant

TABLE IV  
CONNECTIVITY-IMPROVING LINE ADDITIONS THAT DESTABILIZE THE LOW-DAMPED PLL STRESS CASE

| Line  | Graph distance | $\Delta$ Fiedler (%) | $\Delta$ GFV (%) | New real part ( $s^{-1}$ ) | Imag. part ( $s^{-1}$ ) | Interpretation     |
|-------|----------------|----------------------|------------------|----------------------------|-------------------------|--------------------|
| 25–38 | 3              | 0.298                | 0.146            | 0.2502                     | 19.7789                 | strongest reversal |
| 17–29 | 3              | 4.419                | 0.740            | 0.0963                     | 19.8752                 | unstable           |
| 2–29  | 3              | 0.381                | 0.952            | 0.0252                     | 19.6046                 | unstable           |
| 4–30  | 3              | 2.013                | 1.020            | 0.00057                    | 19.7921                 | near boundary      |



(a) Full coupled versus decoupled critical real part.



(b) Standard and generalized-Fiedler connectivity changes.

Fig. 7. IEEE 39-bus line 25–38 stress-case sweep. Connectivity remains above the pre-upgrade case while the full coupled model crosses into instability. The decoupled approximation does not reveal the dangerous mode.

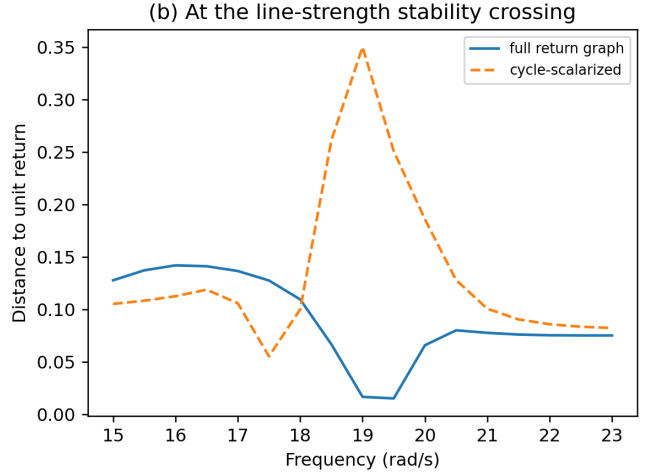
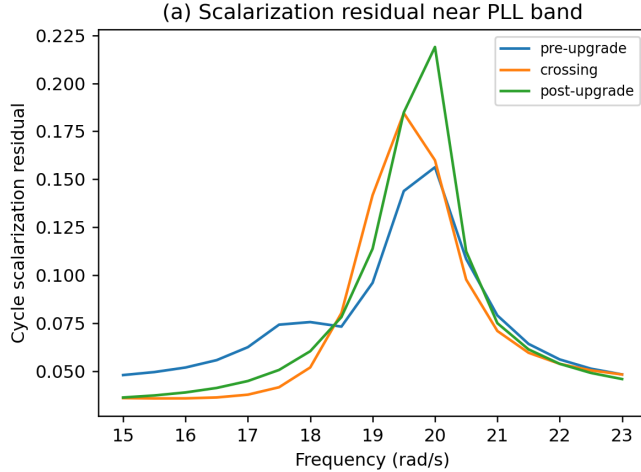


Fig. 8. Cycle-algebra gate for the IEEE 39-bus line 25–38 case. (a) The compatible cycle-scalarization residual grows in the PLL band as the line approaches the stability crossing. (b) At the crossing the full return graph approaches unit return, whereas the cycle-scalarized representation remains far away.

non-Abelian cycle witness, the scalarization residual, and inverse return-ratio margin. The non-Abelian witness develops a local peak in the same approximately 19 rad/s region as return-ratio criticality, while the scalarization residual is broadband and reaches its largest value elsewhere. This separation is important: no individual diagnostic is promoted to a stability proxy; the frequency sweep shows where distinct modeling and resonance mechanisms become active.

The gain-weighted aggregate non-Abelian defect over all chord pairs rises from 0.0452 before the line to 0.0567 at the crossing and 0.0785 after the full line. It becomes 0.0823

after stabilizing retuning; therefore it is *not* a stability margin. Its role is to certify that the coupled cycle algebra cannot be faithfully replaced by independent scalar loops.

Proposition 3 localizes that collective interaction. At the post-upgrade unstable mode the largest gauge-invariant chord closure contributions are shown in Table V. Several converter-related closures contribute simultaneously rather than one isolated “bad cycle.”

Finally, the broadband criterion is not satisfied in the PLL band. For example, the same effective cycle closure associated with the 34–33 port path evaluated at 15 and 25 rad/s has

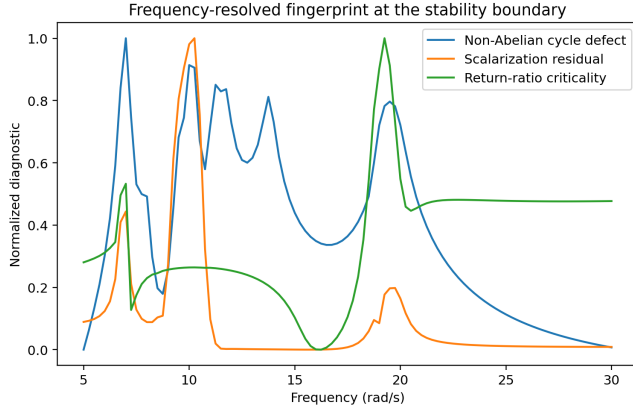


Fig. 9. Frequency-resolved diagnostic fingerprint at the line-strength stability crossing. The three curves are normalized independently; only their frequency localization, not their amplitudes, is compared.

TABLE V  
LARGEST POST-UPGRADE EFFECTIVE CHORD CONTRIBUTIONS

| Source | Target | $ y_i^* H_a x_j $ |
|--------|--------|-------------------|
| 38     | 37     | 0.0822            |
| 30     | 38     | 0.0731            |
| 38     | 30     | 0.0727            |
| 37     | 30     | 0.0501            |
| 34     | 33     | 0.0434            |
| 39     | 30     | 0.0358            |
| 38     | 39     | 0.0273            |
| 39     | 38     | 0.0260            |

a nonzero cross-frequency commutator; a normalized energy-metric witness is approximately 0.71. Hence there is no single common cycle basis that diagonalizes the relevant dynamic transports across that band.

## XI. MODE ATTRIBUTION AND CONTROLLER RETUNING

Biorthogonal participation of the unstable mode is concentrated in the four GFL units, as shown in Fig. 10a. The largest magnitudes are 0.573 at bus 30, 0.380 at bus 37, 0.236 at bus 38, and 0.084 at bus 34. The endpoint bus 38 participates, but the dominant contributor is remote bus 30. This supports the collective-feedback interpretation.

Retuning only the proportional PLL gain at bus 30 gives a zero crossing at

$$K_{p,30}^* = 2.10837. \quad (42)$$

With  $K_{p,30} = 3$ , the new-line case has

$$s_{crit} = -0.10064 + j19.62628 \text{ s}^{-1}, \quad (43)$$

while the improved graph connectivity is unchanged. The practical workflow is therefore

$$\begin{aligned} \text{upgrade} &\rightarrow \text{coupled-mode diagnosis} \\ &\rightarrow \text{device attribution} \rightarrow \text{retuning}. \end{aligned} \quad (44)$$

### A. The reversal is a finite, controller-specific region

The stress result is not an isolated point in parameter space. Figure 11 sweeps the line-strength factor  $\alpha \in [0, 1]$  and a

common GFL PLL proportional gain  $K_p$  for  $K_i = 400$ . The zero contour encloses a finite unstable region. The pre-line stability boundary is  $K_p \approx 0.959$ , whereas with the complete 25–38 line it moves to  $K_p \approx 1.513$ . Thus  $K_p = 1$  is stable before the line and unstable after a sufficiently strong line; even  $K_p = 1.5$  is baseline-stable but gives a small positive post-line real part ( $+0.00629 \text{ s}^{-1}$ ). For  $K_p \geq 2$  the tested GFL model remains stable.

The boundary persists when  $K_i$  is changed. At  $\alpha = 1$ , the minimum stable common  $K_p$  is approximately 1.253, 1.513, and 1.741 for  $K_i = 300, 400$ , and 500, respectively. This establishes a nonzero parameter region rather than a single hand-picked tuning.

A negative control is equally important. Replacing the four converter ports by the tested GFM  $P$ - $f$ / $Q$ - $V$  surrogate gives stable pre/post-line modes for droop values  $R_p = 0.02, 0.04$ , and 0.08, despite the same connectivity increase. Similarly, well-damped GFL tuning ( $K_p = 25$ ) remains stable. The reversal is therefore *controller-specific*; it is not a topological claim that adding line 25–38 is inherently destabilizing.

### B. Penetration and operating-point robustness

We next randomized line additions and converter placements while retaining only baseline-stable systems. For the PLL-dominated GFL family, 120 trials were evaluated at each of six penetrations. No connectivity-improving destabilization was observed from 0–60% GFL penetration; 3/120 trials (2.5%) produced reversals at 80% and another 3/120 at 100%. In contrast, none of 720 matched GFM-placement trials across 0–100% penetration produced a reversal under the tested droop/filter ranges. These frequencies are not estimates of field probability; they are a technology-specific robustness control showing that the effect is neither ubiquitous nor tied only to one four-converter placement.

The same line can also change sign as an intervention when the AC operating point changes. We scaled all loads, resolved the nonlinear power flow, and recomputed the minimum common PLL  $K_p$  required for small-signal stability before and after the complete 25–38 line. At 80% loading the line *reduces* the required  $K_p$  by 0.360 (stabilizing), whereas at 90%, 100%, and 110% loading it *increases* the threshold by approximately 0.148, 0.554, and 0.172, respectively. At 120% loading the reduced stress model is unstable throughout the tested  $K_p \in [0.4, 4]$  interval and is therefore excluded from threshold comparison. Figure 12 summarizes both robustness experiments.

### C. Time-domain consistency

A small localized PLL-angle perturbation at bus 38 was propagated through the linearized system. Over 20 s, the unretuned new-line RMS PLL-angle perturbation grows from approximately  $5 \times 10^{-4}$  to  $4.93 \times 10^{-2}$ , whereas the retuned case decays to approximately  $4.1 \times 10^{-5}$ .

## XII. INTERPRETATION: WHAT IS NEW AND WHAT IS NOT

### A. What the results establish

The experiments and proofs support four statements.

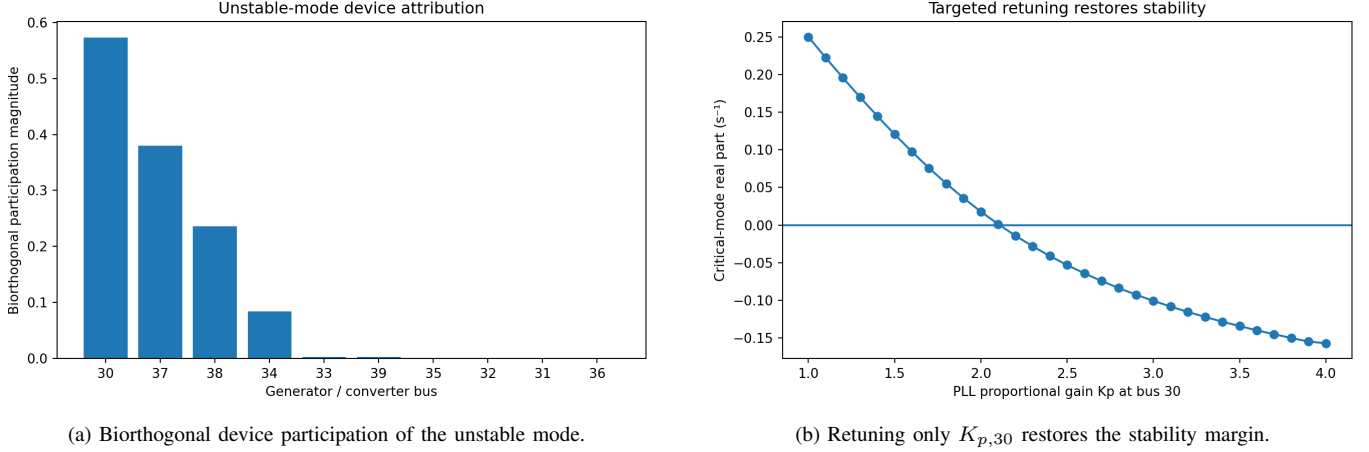


Fig. 10. Diagnosis and mitigation. The line upgrade is retained; only the dominant converter controller is retuned.

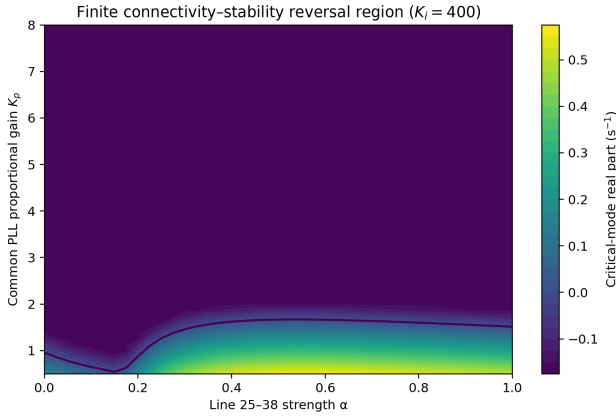


Fig. 11. Finite connectivity–stability reversal region for  $K_i = 400$ . The contour is the full coupled stability boundary. Connectivity remains above the pre-upgrade case as  $\alpha$  grows; the instability is controlled by the closed-loop device/network interaction.

- (1) Exact pointwise scalarization is controlled by simultaneous diagonalizability of fundamental-cycle holonomies, while a common scalar basis over a band requires simultaneous diagonalization of the full frequency-indexed family. Noncommutativity and Jordan defectiveness are distinct obstructions.
- (2) A physically metric-compatible commutator is coordinate invariant, and Theorem 3 turns the commutator into a quantitative off-diagonal error certificate with a spectral-gap/conditioning penalty.
- (3) Global resonance can be collective across cycles. Proposition 3 provides a gauge-invariant chord sensitivity, and the IEEE 39-bus case exhibits several simultaneous converter-related cycle closures rather than one isolated resonant loop.
- (4) Scalar connectivity and the coupled closed-loop spectrum are not ordered in general. The IEEE reversal occupies a finite controller/line-strength region, is absent under nominal GFL tuning and the tested GFM controls, and can be repaired by targeted retuning without undoing the

connectivity upgrade.

### B. What the results do not establish

The non-Abelian defect  $\mathfrak{N}$  is not claimed to be a stability margin or a universally superior empirical classifier. High  $\mathfrak{N}$  means that a scalar representation is structurally questionable, not that the system is necessarily unstable. The fixed-topology screening ablation finds that a simple local  $dq$  coupling statistic has higher average ROC AUC for unsafe-scalarization events; the role of the cycle algebra is instead to provide an exact global reducibility criterion, an error certificate, and topology-only counterexamples that local statistics cannot rule out. The nominal IEEE 39-bus scan found no unstable line addition. The reversal appears in a deliberately low-damped PLL stress case, which is exactly where coupled modeling should matter most.

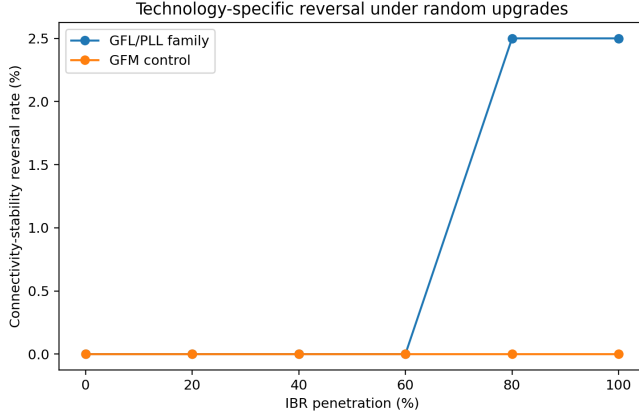
The AC network and operating-point re-solution are benchmark-consistent, but the GFL/GFM dynamic blocks are reduced surrogates. Therefore the IEEE 39-bus result is a constructive AC-consistent proof-of-concept, not yet a full-fidelity industry validation.

### C. Relation to complex frequency

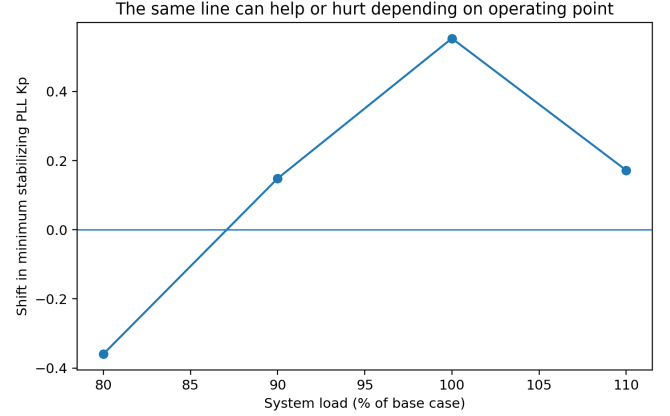
Sofos and Milano showed that complex frequency can coincide with system eigenvalues for diagonalizable LTI systems under the appropriate transformed states, while nonlinear trajectories generally break this equivalence [3]. This gives a natural future observation layer: the coupled mode predicted by the matrix graph may be tracked through complex-frequency patterns measured at converter terminals. That measurement connection is not required for the theorems above and is intentionally left outside the contribution claim.

## XIII. ENGINEERING WORKFLOW

The proposed use of the framework is not to replace full MIMO stability analysis. It is to decide when a reduced scalar analysis is defensible and, when it is not, to preserve the smallest coupled model required for diagnosis.



(a) Random upgrade trials across IBR penetration.



(b) Operating-point dependence of the 25–38 intervention. Positive values mean that the line requires more PLL damping for stability.

Fig. 12. Robustness controls. The connectivity–stability reversal is controller-technology and operating-point dependent rather than an inherent property of a particular line.

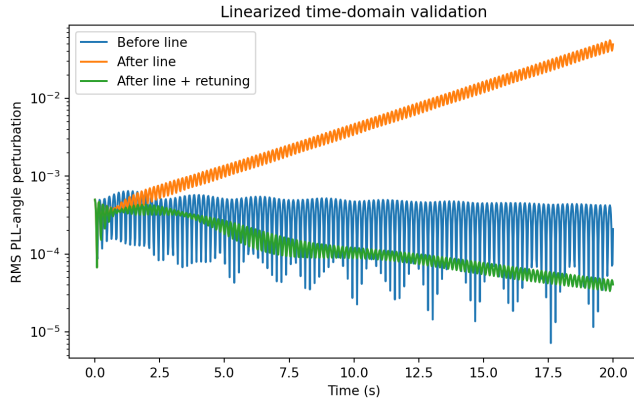


Fig. 13. Linearized time-domain validation of the line 25–38 stress case. The new-line response grows without retuning and decays after targeted PLL retuning.

- Step 1.** Solve the AC operating point and obtain a local port model  $F(s)$ , either analytically, by frequency response, or by DAE elimination.
- Step 2.** Construct the matrix return graph  $T(s)$  and a fundamental-cycle basis.
- Step 3.** Carry a physical port metric, whiten the cycle matrices, and evaluate commutators, Jordan/eigenvector conditioning, and the bound (16) over the frequency band of interest.
- Step 4.** If the holonomy family is diagonalizable and commuting at the required frequency resolution, use the exact or bounded near-scalar channels. If a common broadband basis is required, also test the cross-frequency condition (9); otherwise retain the coupled model.
- Step 5.** Compare graph planning metrics such as GFV with the full coupled spectral margin; do not infer one from the other.
- Step 6.** If an intervention creates a dangerous coupled mode,

use left/right participation and, ultimately, cycle-rooted attribution to identify the controller-network structures that move it.

- Step 7.** Retune the smallest responsible controller set and verify the full coupled and time-domain response.

#### XIV. LIMITATIONS AND FINAL VALIDATION GATE

The remaining external validation gate is one full-order replication. The public EPFL IEEE 39-bus benchmark explicitly provides full-replica Config. I, low-inertia Config. II (four synchronous machines replaced by type-III wind plants), a BESS/VSC variant, and very-low-inertia Config. III [14]. Those models require the MATLAB/Simulink/ARTEMIS environment and cannot be executed in the present reproducibility runtime; alternatively, an independently implemented ANDES/EMT model can be used. The required gate is narrow:

- (i) reproduce a connectivity-improving network intervention in a full DAE/EMT model;
- (ii) compare full MIMO and decoupled analysis;
- (iii) confirm the critical mode in time domain;
- (iv) retune the highest-participation controller and recover the margin without undoing the network upgrade.

A successful full-order reversal would elevate the current proof-of-concept into a strong validation of the proposed screening logic. A failure to reproduce it would still leave the scalarizability theorem and nonordering counterexample intact, while narrowing the engineering claim.

#### XV. CONCLUSION

This paper separates model reducibility from power-system stability. After port reduction, a multi-inverter network becomes a frequency-dependent matrix graph whose fundamental-cycle holonomies contain the global coordinate-invariant obstruction to scalar channels. Exact pointwise and broadband scalarizability reduce to simultaneous diagonalization, with noncommutativity and Jordan defectiveness as separate obstructions. A spectral-gap theorem converts the cycle commutator into a

quantitative bound on the chord off-diagonal residual; in 2,000 structured networks the bound has zero violations and modest slack. A separate 2,000-pair Laplacian-cospectral campaign isolates information that scalar topology and local coupling cannot contain: scalar spectra remain identical to numerical precision while 4.35% of full-MIMO pairs disagree on stability. Global unit-return resonance is also demonstrably collective: the IEEE stress-mode decomposition has an effective closure count of 28.9, and no individual fundamental closure dominates the mode.

The IEEE 39-bus reduced study then supplies a deliberately falsification-oriented engineering example. Connectivity improves under line 25–38, yet a low-damped GFL region contains a finite parameter set in which the full coupled mode crosses into instability while a decoupled model misses the event. Frequency-resolved cycle diagnostics show a local non-Abelian feature near the critical return-ratio band while the reduction error remains broadband; penetration and loading sweeps show that the effect is technology- and operating-point-dependent rather than ubiquitous. The new cycle diagnostics show that the same mode cannot be reconstructed by independent scalar cycle channels, and chord sensitivity exposes a distributed converter-related feedback structure. Well-damped GFL tuning and the tested GFM controls remain stable, while targeted PLL retuning repairs the stress case without removing the line. The conclusion is therefore narrower than a new strength metric: *non-Abelian cycle algebra certifies when scalar analysis is structurally inadequate; stability remains a property of the full closed-loop matrix graph.*

#### APPENDIX A

##### BROADBAND AND JORDAN DETAILS

Theorem 2 is a statement about one common similarity on the root fiber after frequency-dependent tree normalization. It should not be confused with a frequency-independent local state transformation in the original node coordinates. The distinction matters because the tree transport itself can be dispersive. In computations, a sampled broadband obstruction can be formed from cross-frequency commutators of the whitened holonomy family. A zero commutator family must still be checked for diagonalizability; the exact deficiency (11) and eigenvector conditioning provide complementary tests.

#### APPENDIX B

##### PROOF DETAILS FOR THEOREM 3

Let  $B_q = S^{-1}\hat{H}_q S$ . Since  $\Lambda$  is diagonal,

$$\|[\Lambda, B_q]\|_F^2 = \sum_{i \neq j} |\lambda_i - \lambda_j|^2 |(B_q)_{ij}|^2 \geq \delta_0^2 \|\text{off}(B_q)\|_F^2. \quad (45)$$

Moreover,

$$[\Lambda, B_q] = S^{-1}[\hat{H}_0, \hat{H}_q]S, \quad (46)$$

so  $\|[\Lambda, B_q]\|_F \leq \|S^{-1}\|_2 \|S\|_2 \|\hat{H}_0, \hat{H}_q\|_F$ . Combining the inequalities proves (16). The whitening construction is gauge compatible: if  $G = W^*W$  and  $H' = Q^{-1}HQ$ ,  $G' = Q^*GQ$ , choosing  $W' = WQ$  yields  $W'H'(W')^{-1} = WHW^{-1}$ .

#### APPENDIX C

##### ADDITIONAL PROOF DETAILS FOR THEOREM 1

Let the spanning tree be rooted at  $r$ . Set  $Q_r = I$ . For every tree edge  $i \rightarrow j$ , define  $Q_j = T_{ji}Q_i$ ; if the stored orientation is opposite to the tree traversal, use the inverse transport. Then

$$Q_j^{-1}T_{ji}Q_i = I. \quad (47)$$

Every chord  $a : u \rightarrow v$  becomes  $Q_v^{-1}T_aQ_u$ , which is the holonomy obtained by traversing the tree from  $r$  to  $u$ , then the chord, then the inverse tree path from  $v$  to  $r$ . If a second gauge  $S_i$  leaves every tree edge equal to  $I$ , then  $S_j^{-1}S_i = I$  on every tree edge; connectedness implies  $S_i = S$  for all  $i$ . Consequently all chords can be made diagonal exactly when one common  $S$  diagonalizes them. This proves the equivalence in Theorem 1.

#### APPENDIX D

##### COUNTEREXAMPLE STABILITY THRESHOLD

For (25), the relative-mode trace is

$$\text{tr}(A - \mu C) = \text{tr}(A) - \mu \text{tr}(C), \quad (48)$$

which remains negative at the first positive determinant root. Hence the two-dimensional Routh–Hurwitz conditions reduce the first loss of stability to  $\det(A - \mu C) = 0$ , yielding (28). This makes the counterexample exact rather than simulation-dependent.

#### APPENDIX E

##### COLLECTIVE-CYCLE COUNTEREXAMPLE DATA

For Proposition 2, the two chord holonomies in tree gauge are

$$\begin{aligned} H_1 &= \begin{bmatrix} 0.455247 & 0.113039 \\ 0.116163 & 0.312369 \end{bmatrix}, \\ H_2 &= \begin{bmatrix} 0.498997 & 0.057555 \\ 0.017625 & 0.188825 \end{bmatrix}. \end{aligned} \quad (49)$$

Their energy-metric commutator defect for  $G = I$  is 0.08588. The full  $8 \times 8$  return operator is scaled so that its dominant eigenvalue is exactly one; the complete matrix is included in the reproducibility package. The important point is structural: the global unit-return condition emerges without either  $H_1$  or  $H_2$  approaching one individually.

#### APPENDIX F

##### REDUCED DEVICE-MODEL SCOPE

The IEEE 39-bus surrogate intentionally retains only the dynamic mechanisms required to test the paper hypothesis. SG inertia/damping is represented explicitly; GFL devices retain PLL states and the operating-point rotation that mixes active/reactive injection channels; line additions trigger a new nonlinear AC solution before each linearization. Current limits, dc-link energy, detailed electromagnetic filters, saturation, protection logic, and vendor-specific inner loops are omitted. These omissions define the final full-order validation gate rather than being hidden assumptions.

## APPENDIX G

### REPRODUCIBILITY SUMMARY

The numerical package accompanying this draft contains: (i) the 2,000-case scalarizability-bound campaign; (ii) the 2,000-pair Laplacian-cospectral topology-only campaign and exact iso-local counterexample; (iii) the 1,000-case IEEE screening ablation; (iv) the analytical connectivity counterexample; (v) the MATPOWER case39 data and AC power-flow/Schur-reduction SG/GFL/GFM surrogate; (vi) frequency-resolved cycle diagnostics and exact unit-return closure decomposition; (vii) penetration, loading, line-strength, controller-retuning, and time-response sweeps; and (viii) all figure data. The principal random seed is 20260823.

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